POWER: USB-C 5V (D101) / LiPo (Q101) auto power path -> VSYS -> U6 LP5907 3.0 V -> +3V0_A (ADS1298 AVDD)
and U9 AP2112K 3.3 V -> +3V3 (ESP32-S3, ADS1298 DVDD). U6 EN = +3V3, so the AFE is off whenever the MCU rail is off.

B5819W
D101
VBUS
VSYS
#FLG07
PWR_FLAG
VSYS
AO3401A
Q101
LP5907MFX-3.0
U6
VSYS 1
+3V3 3
IN
EN
OUT
GND
5 +3V0_A
AGND 2
+3V0_A
C101 10nF
AGND
+3V0_A
C102 100nF
AGND
R108 100k
VBUS
AGND
VBUS 1
VBAT 3
VSYS 2
C107 10uF
VSYS
AGND

TEST POINTS
+3V0_A TP
+3V3 TP
AGND TP
VSYS TP
RLD_OUT TP
AGND TP

LiPo pads (protected)
J3
1 VBAT
2 AGND
LiPo: J3 = two solder pads for a protected LiPo. U8 MCP73831 charges it from USB at 100 mA (R_PROG 10k).
#FLG02
PWR_FLAG
AGND
C120 100nF +3V0_A AGND
C119 100nF +3V0_A AGND
C118 10uF +3V0_A AGND
C121 10uF +3V3 AGND
C122 100nF +3V3 AGND

ADS1298IPAGR
U1
RLD: 1M || 1.5nF feedback, 330k series to the REF electrode(s) limits DC fault current to <10 uA.
R101 1M RLD_INV RLD_OUT
C111 1.5nF RLD_INV RLD_OUT
R102 330k RLD_OUT RLD_ELEC

INPUT RC: 22k series (limits fault and plug-insertion current) + 1nF C0G to AGND, ~7 kHz.
ADS1298 oversamples at 512 kHz, so this is the only anti-alias filter needed.

CH1 (TRS)
J1
S RLD_ELEC
R ELEC1_N
T ELEC1_P

ELEC1_P R201 220k CH1_P C201 1nF C0G AGND
ELEC1_N R202 220k CH1_N C202 1nF C0G AGND
ELEC2_P R203 220k CH2_P C203 1nF C0G AGND
ELEC2_N R204 220k CH2_N C204 1nF C0G AGND
ELEC3_P R205 220k CH3_P C205 1nF C0G AGND
ELEC3_N R206 220k CH3_N C206 1nF C0G AGND

CH1_P 16 IN1P
CH1_N 15 IN1N
CH2_P 14 IN2P
CH2_N 13 IN2N
CH3_P 12 IN3P
CH3_N 11 IN3N
+3V0_A 10 IN4P
+3V0_A 9 IN4N
+3V0_A 8 IN5P
+3V0_A 7 IN5N
+3V0_A 6 IN6P
+3V0_A 5 IN6N
+3V0_A 4 IN7P
+3V0_A 3 IN7N
+3V0_A 2 IN8P
+3V0_A 1 IN8N
VCAP1 28 VCAP1
VCAP2 30 VCAP2
VCAP3 55 VCAP3
VCAP4 26 VCAP4
VREFP 24 VREFP
AGND 25 VREFN
AVDD 19 +3V0_A
AVDD1 54 +3V0_A
DVDD 48 +3V3
AVSS 29 AGND
AVSS1 53 AGND
DGND 33 AGND
WCT 64
RLDOUT 63 RLD_OUT
RLDIN 62 RLD_ELEC
RLDREF 61 RLD_INV
60
TESTP_PACE_OUT1 17 +3V0_A
TESTP_PACE_OUT2 18 +3V0_A
GPIO1 52
GPIO2 51
GPIO3 50
GPIO4 49
41 AGND
DAISY_IN 40 SPI_SCK
SCLK 39 SPI_MOSI
DIN 38 ADS_CS
CS 43 SPI_MISO
DOUT
CLK 37
CLKSEL 42 +3V3
START 58 ADS_START
DRDY 57 ADS_DRDY
PWDN 56 ADS_PWDN
RESET 59 ADS_RESET
RESV1 31 AGND

VREFP
C112 10uF AGND
C113 100nF AGND
VCAP1 C114 22uF AGND
VCAP2 C115 1uF AGND
VCAP3 C116 1uF AGND
VCAP4 C117 1uF AGND

ELECTRODES: J1 TRS for CH1 (T +, R −, S REF).
J6 1 mm FPC carries CH1–CH3 for a flex array or breakout,
in parallel with J1 for CH1: use one or the other.
All REF pins share RLD_ELEC. J6 1/5/9 + MP = AGND guards.
FLEX ARRAY CH1–3 (FPC)
J6
AGND 1
ELEC1_P 2
ELEC1_N 3
RLD_ELEC 4
AGND 5
ELEC2_P 6
ELEC2_N 7
RLD_ELEC 8
AGND 9
ELEC3_P 10
ELEC3_N 11
RLD_ELEC 12
AGND_MP

SAFETY: with electrodes on the body and USB-C plugged in, run the laptop on battery,
unplugged from mains, or use a simulated input. Charge while worn only from
an isolated (Class II) charger.
LAYOUT: keep U1 and its supply away from the XIAO antenna.

ADS_CS R103 10k +3V3
ADS_RESET R105 10k +3V3
ADS_PWDN R106 10k +3V3

ADS1298: CH1–CH3 used, CH4–CH8 and TESTP tied to AVDD per datasheet (power down in firmware).
1, 2 or 3 channels by mode, no rework. HPF/LPF in firmware: capture ~0.5 Hz up at 4 kSPS+.
Lead-off detection and RLD channel selection are firmware settings.
Internal clock (CLKSEL high), internal 2.4 V reference.

MCU: ESP32-S3-MINI-1-N8 (PCB antenna, keep copper out of the antenna end). Native USB on IO19/IO20 through USBLC6 ESD to USB-C.
ADS1298 SPI: IO12 SCLK, IO13 DOUT (MISO), IO14 DIN (MOSI), IO15 CS, IO16 DRDY, IO17 START, IO18 RESET.
Flashing over USB uses the built-in USB-Serial-JTAG. If it ever stops enumerating: short TP10 (IO0) to GND, then tap TP9 (EN) to GND.
Charger U8 MCP73831 at 100 mA (R703 10k). 3.3 V rail U9 AP2112K from VSYS.

USB-C
J7
VBUS A4 VBUS
CC1 A5 USB_CC1
CC2 B5 USB_CC2
D- A7 USB_DN
D- B7 USB_DN
D+ A6 USB_DP
D+ B6 USB_DP
SBU1 A8
SBU2 B8
SHIELD AGND_SH
GND A1 B12
AGND
#FLG08
PWR_FLAG
VBUS
R701 5.1k USB_CC1 AGND
R702 5.1k USB_CC2 AGND
U7
USBLC6-2SC6
USB_DP 1 6
USB_DN 3 4
2 5
VBUS
AGND

ESP32-S3-MINI-1-N8
U2
3V3 3 +3V3
EN 49
GND 1 AGND
TXD0 39
RXD0 40
ESP_BOOT 4 IO0
IO1 5
IO2 6
IO3 7
IO4 8
IO5 9
IO6 10
IO7 11
IO8 12
IO9 13
IO10 14
IO11 15
IO12 16 SPI_SCK
IO13 17 SPI_MISO
IO14 18 SPI_MOSI
IO15 19 ADS_CS
IO16 20 ADS_DRDY
IO17 21 ADS_START
IO18 22 ADS_RESET
USB_D- 23 USB_DN
USB_D+ 24 USB_DP
IO21 25
IO26 28
IO33 29
IO34 30
IO35 31
IO36 32
IO37 33
IO38 34
IO39 35
IO40 36
IO41 37
IO42 41
IO45 27
IO46 26
IO47 42
IO48 43
GND2 44
GND3 45
GND4 46
GND5 47
GND6 48
GND7 50
GND8 51
GND9 52
GND10 53

U8
MCP73831T-2ACI/OT
VBUS 1
VDD
VBAT 3
PROG 5
VSS 2
STAT 1
AGND 2
C104 4.7uF VBUS AGND
C105 4.7uF VBAT AGND
CHG_PROG 5
R704 10k CHG_PROG AGND

R709 10k +3V3 ESP_EN
C108 1uF ESP_EN AGND
TP9 EN (reset) ESP_EN
TP10 BOOT (IO0) ESP_BOOT

U9
AP2112K-3.3TRG1
VSYS 1
VIN
EN 3
VOUT 5 +3V3
GND 2 AGND
C109 1uF VSYS AGND
C110 10uF +3V3 AGND
C111 100nF +3V3 AGND

Spec: emg_headband/hardware/emg_afe/README.md
1 to 3 channel surface EMG, ADS1298 + ESP32-S3-MINI-1, JLCPCB assembly

Sheet: /
File: emg_wearable_revA.kicad_sch

Title: EMG wearable AFE
Size: A2 | Date: 2026-10-11 | Rev: A
KiCad E.D.A. 10.0.7 | Id: 1/1